***Question 1**:* Arguments can be constructed by chaining facts, and logic programming rules that can include in their antecedents expressions of the form `NOT X' which is read as `X is not provable'. Then, an argument A attacks an argument B if A claims X and B is constructed using a rule that assumes X is not provable. Consider the facts and rules below:

Facts	**Rules**	
p	Rule1 =	j → g
s	Rule2 =	g and p and NOT q → r
j	Rule 3 =	t and NOT s → q
t	Rule 4 =	w and NOT z and NOT r → s
m	Rule 5 =	k and NOT v → z
k	Rule 6 =	m → v
i	Rule 7 =	i → w

The facts j and p together with rules 1 and 2, can be used to construct the argument X, written as [j, j → g, p, g and p and NOT q → r] : r. The fact t and rule 3 can be used to construct Y that claims q and so attacks X. Note also that arguments can also consist of just facts. So [p] : p, [s] : s, …. are also arguments.

Write out the argument framework that can be constructed from the belief base of facts and rules in the table above. That is to say, write out the arguments that can be constructed (you can use any upper case letters to name the arguments) and the attacks amongst the constructed arguments.

Then state what the preferred labellings and the grounded labelling of the framework are.

Arguments:

A = [p] : p, B = [s] : s, C = [j] : j, D = [t] : t, E = [m] : m, F = [k] : k, G = [i] : i

H = [j; j → g] : g

I = [j, j → g; p; g and p and NOT q → r] : r

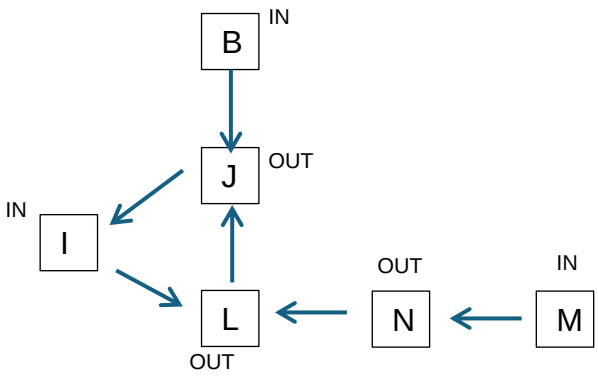
J = [t; t and NOT s → q] : q

K = [i ; i → w] : w

L = [i ; i → w ; w and NOT z and NOT r → s] : s

M = [m ; m → v] : v

N = [k ; k and NOT v → z] : z



Question 2: Consider the following stratified KB of classical formulas:

$$\neg a \vee \neg b \vee \neg c$$

$$a \qquad a \rightarrow d$$

$$b \qquad c \qquad \neg d \vee \neg c$$

What are the preferred maximal consistent subsets?

Answer:

$$\{\neg a \vee \neg b \vee \neg c, \quad a, \quad a \rightarrow d, \quad b, \quad \neg d \vee \neg c\}$$
$$\{\neg a \vee \neg b \vee \neg c, \quad a, \quad a \rightarrow d, \quad c\}$$

How would you define a default logic theory such that the credulous and sceptical nm consequences are the same as those defined by the stratified KB (hint: you no longer need assume a *total* linear ordering over the defaults; relax this assumption so as to assume a *partial* linear ordering over the defaults **and** if a default *i* can be applied, only apply it if there are no applicable defaults *j,k,l…* such that *j > i, k >i, l > i… and applying j,k,l…* and *i* would result in inconsistency).

Answer:

1. $\top \Rightarrow \neg a \vee \neg b \vee \neg c$ 2. $\top \Rightarrow a$ 3. $\top \Rightarrow a \rightarrow d$

4. $\top \Rightarrow b$ 5. $\top \Rightarrow c$ 6. $\top \Rightarrow \neg d \vee \neg c$

1 > 2, 1> 3, 2 > 4, 2 > 5, 2 > 6, 3 > 4, 3 > 5, 3 > 6

Suppose you start by considering whether you can apply 5 and then 6. Apply 5 to get *c*. You cannot then apply 6 as this would result in an inconsistent outcome if one were to apply defaults that are ordered above 6 (i.e. 2 and 3). So this rules out any extension with 5 and 6.

You should get the same results as you would if you were to complete the partial linear ordering into all possible total orderings (as I described in the LGT). For each total completion, calculate all extensions. Then define credulous/sceptical consequences in the usual way.